Venkata Lakshmi Parimala Pasupuleti

Quantitative Model Validation Analyst · Python · Regression Modeling · Statistical Testing · Financial Data · SQL · Model Risk

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PROFESSIONAL SUMMARY

Quantitative analyst with 7+ years working with large-scale financial datasets and a strong foundation in **regression methodologies, statistical diagnostic tests, and model validation** at production scale. At Intuit, builds and executes **model validation frameworks** for financial analytics AI — assessing model conceptual soundness, designing benchmarking and **outcome analysis** suites, documenting findings with structured audit trails for stakeholder review. At Entrata, processed and validated **1.6M+ HUD/Tax Credit financial records** (income certifications, asset disclosures, rent-to-income ratios) with 100% PHI compliance. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026) — statistical modeling, regression analysis, anomaly detection, data governance. CrashClarity (Random Forest + Logistic Regression, 85%+ accuracy, 104K records) and Real Estate Analysis (**XGBoost, $R^2 = 0.977$**) demonstrate hands-on **regression methodology and sensitivity analysis** depth. Python (9+ yrs), SQL, scikit-learn, XGBoost; experienced leveraging **AI tools and agents** to accelerate validation workflows.

TECHNICAL SKILLS

Quantitative & Model Validation: Regression Methodologies, Statistical Diagnostic Tests, Model Conceptual Soundness, Benchmark Testing, Sensitivity Analysis, Outcome Analysis, Model Documentation & Reporting, Risk Rating Analysis, Financial Data Validation

Analytics & Modeling: Python (9+ yrs), SQL, R, SAS, scikit-learn, XGBoost, Random Forest, Logistic Regression, Pandas, NumPy, Feature Engineering, Cross-Validation, Statistical Modeling

AI Tools & Agents: LLM Evaluation, LangGraph, LangChain, RAG, AI Agents, Anthropic Claude, OpenAI APIs, HuggingFace, FastAPI

Cloud & Data Infrastructure: AWS (S3, EC2, RDS, Glue), Snowflake, PostgreSQL, MySQL, SQLite, ETL Pipelines, Data Quality Monitoring, Docker, GitHub Actions CI/CD

WORK EXPERIENCE

Intuit

Jun 2026 – Present

Software Engineer — Model Validation & ML Pipelines · Financial Analytics

- **Build and execute model validation frameworks** for financial analytics AI — regression testing suites (golden datasets + benchmarking) assessing **model conceptual soundness**, detecting performance degradation and model drift; design **sensitivity analysis and outcome analysis** to evaluate model limitations and risk; document and communicate validation findings to business stakeholders with structured audit trails ensuring risks are accurately identified and escalated.
- Build end-to-end ML pipelines (scikit-learn, XGBoost) on Snowflake and AWS from data ingestion through feature engineering, model evaluation, and prediction APIs; **leverage AI tools and agents** to accelerate coding, research, and validation workflows; ship via GitHub Actions CI/CD with logging for production visibility.

Entrata — Multifamily & Affordable Housing SaaS

Jan 2017 – Jul 2024

Analytic Software Engineer — Financial Data Validation & Compliance Analytics

- Validated and processed **1.6M+ HUD/Tax Credit financial records** (income certifications, asset disclosures, rent-to-income ratios, recertification data) — assessed data accuracy, flagged income documentation inconsistencies, documented findings for compliance reporting; maintained **100% PHI compliance with zero violations**; managed data access governance via AWS S3 Libsodium-encrypted pipelines with role-based access controls.
- Automated compliance review workflows across 1M+ records, eliminating **80% of manual review workload** with zero compliance violations; optimized **200+ SQL queries (30% performance improvement)**; delivered across **120+ Agile sprints**.

PROJECTS

CrashClarity — Regression Modeling on 104K Records with Self-Service Dashboard

Regression · Statistical Modeling

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages — Live Dashboard

- Built Random Forest + Logistic Regression ensemble with feature engineering on 104K+ UK accident records; applied **regression methodologies, statistical diagnostic tests, and outcome analysis** achieving **85%+ classification accuracy**; deployed interactive self-service dashboard translating model findings to strategic insights.

Real Estate Price Analysis: Connecticut Property Market (2001–2022)

ML · Sensitivity Analysis

Python · XGBoost · Random Forest · Pandas · Matplotlib · Seaborn · Feature Engineering

- XGBoost and Random Forest models on assessed value, sale ratio, property type, and geographic features achieving **$R^2 = 0.977$**; conducted **sensitivity analysis** across income brackets and property classes; documented model assumptions, limitations, and risk drivers across two decades of housing data.

Leasel — Financial Data Validation & Compliance Audit Pipeline

Model Validation · Audit

Python · LangGraph · scikit-learn · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- End-to-end **model validation and governance pipeline** across 1,240 unstructured lease documents — feature extraction, rule-based classification, deterministic audit engine with **full metadata and lineage**; **93.6% precision, 87.9% F1** across 10 compliance rules; every determination traceable, documented, and audit-ready.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Statistical modeling, regression analysis, data governance, ML pipelines, anomaly detection

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017